

YASH KUMAR SINGH

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B.Tech CSE | VIT Bhopal University | 2023–2027

TECHNICAL SKILLS

Programming: Python

Libraries: Pandas, NumPy, Scikit-learn, OpenCV, YOLO

EDUCATION

VIT Bhopal University, Bhopal, Madhya Pradesh

2023 – 2027

B.Tech in Computer Science and Engineering | CGPA: 7.1/10

Expected May 2027

Lucknow International Public School, Lucknow, Uttar Pradesh

Jul 2022

12th Standard | CBSE GPA: 7.6/10

Lucknow International Public School, Lucknow, Uttar Pradesh

Jul 2020

10th Standard | CBSE GPA: 7.4/10

EXPERIENCE

Tata Motors Ltd. — Intern

Jul 2026 – Present

- Currently working as an Intern at Tata Motors, gaining practical industry exposure and professional experience.

PROJECTS

CCTV Video Analysis | Python, YOLO, OpenCV

- Developed a CCTV video analysis system aimed at improving security monitoring for large farmhouses and other properties.
- Designed the project to help property owners review surveillance footage and understand activities that occurred during unattended periods, particularly at night.
- Used Python and OpenCV to process and analyze recorded CCTV video footage.
- Processed surveillance footage to support systematic analysis of activities captured by CCTV cameras.
- Applied object detection techniques to identify relevant persons/objects appearing in video frames.
- Applied object tracking techniques to analyze the movement of identified persons/objects across consecutive video frames.
- Enabled systematic analysis of recorded footage to help identify and review security-relevant activities.
- Focused on reducing the effort required to manually examine long CCTV recordings and making surveillance review more efficient.

Car Price Prediction | Python, Pandas, NumPy, Scikit-learn

- Developed a machine learning model to predict used-car prices from vehicle specifications.
- Used **company, car name, model type, fuel type, and kilometers driven** as input features for price prediction.
- Used historical vehicle data with **selling price as the target variable** for model training.
- Performed data cleaning and preprocessing to prepare the vehicle dataset for analysis and model training.
- Applied **categorical encoding** to convert non-numeric vehicle attributes into numerical values.
- Performed **exploratory data analysis (EDA)** using Pandas and NumPy to understand patterns and relationships in the dataset.
- Trained a **regression model using Scikit-learn** to estimate used-car prices.
- Evaluated the trained model to assess its ability to generate price predictions for vehicle data.

LEADERSHIP & EXTRA-CURRICULAR

Defence Warrior Club — Vice President

- Led and coordinated defence-awareness initiatives, student activities, awareness sessions, and outreach for defence examinations such as CDS.
- Coordinated with club members, students, and university stakeholders to plan and execute events and promote awareness of defence careers, discipline, leadership, and fitness.

UI/UX Club — Event Management Team

- Member of the Event Management Team, supporting the planning and coordination of club events and activities.

State-Level Sports Winner

- Won the **Uttar Pradesh State-Level Sports Competition (2019–20)** during Class 10, demonstrating competitive performance and sportsmanship